

THE SMALL INTESTINE (Surgery)

Anatomy & physiology

- **Functions:** nutrient digestion/absorption (carbs, protein, fat, **B12**, minerals); GALT (immune); endocrine (**secretin, somatostatin, CCK, motilin, peptide YY, GLP, neurotensin**); barrier (brush border, tight junctions, **defensins**); enterohepatic bile circulation; intestinal adaptation.
- Landmarks: DJ flexure (**ligament of Treitz**), ileocaecal valve, **plicae circulares (valvulae conniventes)**, **Peyer's patches**.
- **Motility:** intrinsic (**Cajal cells = pacemaker**) + external neuro-hormonal.

Diverticular disease

Meckel's diverticulum

- **Most prevalent congenital GI anomaly. "Rule of 2s":** 2% population, ~2 inches (3–5cm), ~2 feet (60cm) from ileocaecal valve, M:F equal, often <2yrs. **60% heterotopic mucosa (60% gastric)**. From persistent **vitelline (omphalomesenteric) duct** (8th week).
- **Presentation (4–6% complications):** asymptomatic; **GI bleeding (children, painless, maroon stool)**; diverticulitis (mimics appendicitis); peptic ulceration; obstruction (intussusception, volvulus, band); **Littre's hernia** (in inguinal/femoral hernia); carcinoid.
- **Diagnosis:** incidental, barium, **Tc-99m scan (if gastric mucosa)**. **Treatment: diverticulectomy if symptomatic** (asymptomatic = controversial).

Duodenal diverticulum

- Mostly **periampullary (juxtapapillary)**; ↑with age, rare <40; acquired. Causes **cholangitis, choledocholithiasis, chronic pancreatitis, Oddi dysfunction**; problem during ERCP. Treat: asymptomatic → leave; bleeding → surgery.

Jejunal diverticulum

- Multiple; **connective tissue disease**. **Malabsorption, anaemia, steatorrhoea, B12 deficiency**; acute abdomen if inflamed.

Blind loop syndrome

- Stasis → abnormal bacterial flora → **malabsorption (fat + B12), anaemia, steatorrhoea, weight loss**. Causes: long afferent loop (Polya gastrectomy), jejunal diverticula, stricture.

Inflammatory bowel disease (Crohn's focus)

- UC, **Crohn's (regional enteritis)**, indeterminate (15%).
- **Etiology:** unknown; geography, smoking (↑Crohn's), OCP, family history 10–30%, autoimmune + mucosal barrier defect, ?Mycobacterium paratuberculosis.
- **Extra-intestinal: sclerosing cholangitis (40–60%), anaemia, DVT, arthropathy (sacroiliitis, ankylosing spondylitis), ocular (iritis, uveitis, episcleritis), skin (erythema nodosum, pyoderma gangrenosum), finger clubbing, renal stones.**

Crohn's disease

- **F>M; age 15–30 + 55–60; mouth→anus, terminal ileum 60%, rectal sparing 40%. Smoking ↑ risk 3x.**
- **Pathology: non-caseating granulomatous, transmural;** deep serpiginous ulcers, **skip lesions + rectal sparing, cobblestone**, adhesions/fibrosis/stricture/fistula, mesenteric thickening.

- **Clinical:** acute (mimics appendicitis, diarrhoea precedes pain); chronic (mild diarrhoea, RIF pain + mass, **food fear, weight loss**); perianal disease (abscess, fistula); growth retardation in children.
- **Investigations:** sigmoidoscopy/colonoscopy (aphthous ulcers, skip), **barium = "string sign of Kantor"**, CT (fistula/abscess), **MRI (perianal)**.
- **Treatment:** medical (**steroid + 5-ASA**, antibiotics if abscess, **azathioprine = steroid-sparing**, nutrition). **Surgery doesn't cure — for complications only:** obstruction (35%), bleeding, perforation, failed medical, fistula/abscess (30–38%), fulminant colitis, malignancy, perianal. **Stricturoplasty**; segmental/ileocaecal resection.

Intestinal infections

- **Amoebiasis (*Entamoeba histolytica*):** **bottlenecked ulcer (lower sigmoid/upper rectum)**; dysentery, amoebic mass (mimics cancer). **Metronidazole + diloxanide furoate**.
- **Typhoid/paratyphoid:** lower ileum ulcer (longitudinal); complications = ileus, **haemorrhage, perforation**.
- **Intestinal TB:** ileum + proximal colon. **Ulcerative** (secondary to pulmonary TB, transverse ulcers) vs **hyperplastic** (ileocaecal, narrowing — unlike Crohn's rarely fistula). Barium = filling defect terminal ileum. **Chemotherapy**; surgery if obstruction.
- **Actinomycosis:** after appendectomy; **sulphur granules**; penicillin.

Other

- **Pneumatosis cystoides intestinalis:** gas cysts in bowel wall; 15% idiopathic, 85% associated (COPD, asthma). Treat: high O₂, metronidazole.
- **Traumatic rupture:** blunt/penetrating; laparotomy; small clean → primary closure, damaged → resection.
- **Small bowel perforation:** iatrogenic (ERCP) commonest; TB, typhoid, Crohn's, ischaemia, neoplasm, NSAIDs.

Intestinal fistula

- Abnormal communication between epithelialized surfaces. Internal vs external (enterocutaneous). **Low <200 mL/day, high >500 mL/day. >80% iatrogenic** (anastomotic dehiscence).
- **90% close spontaneously in 5 weeks** (10–15% mortality from sepsis).
- **FRIEND (factors preventing closure):** Foreign body, Radiation, Inflammation (IBD), Epithelialization, Neoplasm, Distal obstruction.
- **Therapy:** stabilization (IVF, nutrition, antibiotics, skin protection, drainage, somatostatin) → investigate → definitive surgery (2–3 months).

Small bowel tumours

- **Rare** (only 1.1–2.4% of GI malignancies despite 90% of mucosa) — dilution of carcinogens, rapid transit, low bacteria, IgA, apoptosis. More common in duodenum.
- **Benign: adenoma (commonest)**, lipoma, GIST. Symptomatic → obstruction, bleeding, intussusception.
- **Peutz-Jeghers syndrome:** familial **hamartomatous polyposis (jejunum) + mucocutaneous melanosis (lips/oral)**; bleeding, intussusception; rarely malignant.
- **Malignant:**
- **Adenocarcinoma** (35–50% of small bowel cancer): 5th–6th decade; risk = Crohn's, coeliac, Peutz-Jeghers, FAP, HNPCC. Treat: resection.

- **Carcinoid** (20–40%): from enterochromaffin cells (appendix, ileum); produces **serotonin** → **5-HIAA in urine**. **Carcinoid crisis** (only with metastases): flushing, diarrhoea, hypotension, **right heart valve fibrosis (pulmonary/tricuspid)**. Treat: resection + lymphadenectomy if >2cm; **somatostatin** for symptoms.
- **Lymphoma** (10–15%): Western (B-cell), **Mediterranean (alpha-chain)**, coeliac-associated (T-cell).
- **GIST: KIT (CD117) mutation**, from Cajal cells; mostly stomach; resection; **imatinib (tyrosine kinase inhibitor)** for metastatic/unresectable.

High-yield one-liners

- **Meckel's** = rule of 2s, 60% gastric mucosa, Tc-99m scan, painless GI bleed in children; **Littre's** hernia.
- **Crohn's** = transmural non-caseating granuloma, skip lesions, cobblestone, string sign; surgery for complications only; **FRIEND** for fistula.
- **Amoebiasis** = bottlenecked ulcer + metronidazole; **TB** = ulcerative vs hyperplastic.
- **Small bowel tumours** rare; adenoma commonest benign; **Peutz-Jeghers** = hamartoma + mucocutaneous melanosis.
- **Carcinoid** = serotonin/5-HIAA, crisis with metastases (right heart fibrosis); **GIST** = KIT + imatinib.